

Loan Risk & Lending Performance Analysis

Tools: SQL (MySQL) · Power BI · Data Modeling

Business Problem

Banks need to monitor loan performance and identify high-risk borrowers to reduce defaults and improve profitability. This project analyzes lending data to uncover patterns in borrower risk, loan performance, and repayment behavior — enabling better credit decision-making.

Dataset

- **38,576** loan records · Full year 2021
- **\$435.8M** total funded · **\$473.0M** total received
- **13.8%** charge-off rate — 5,333 defaulted loans
- **\$28.2M** permanent net loss — \$65.5M lent to defaulters, \$37.3M recovered (**56.9%** recovery rate)

Key Insights

- **Grade B & C are the real loss driver, not Grade G.** They account for 49% of all defaults — high volume at moderate risk outweighs low volume at extreme risk.
- **60-month loans default at 2.1× the rate of 36-month loans.** All 1,098 currently open loans are 60-month and still developing; average loan size is 1.6× larger, amplifying dollar loss per default.
- **Business-purpose loans carry structural risk beyond credit grade.** Default rate is 11.4% even at Grade A — 2× all other purposes at the same grade.
- **Largest dollar risk: Grade E + Debt Management — \$31.8M at 24.8% default rate.** Invisible from individual charts; surfaces only via cross-filtering the detail page.
- **Verified borrowers default more than unverified ones (15.7% vs 12.2%).** Verified borrowers receive ~2× larger loans on average — verification flags risk but doesn't limit exposure.

Business Impact

- Reframed risk focus — Grade B & C concentration is a bigger priority than high-grade outliers
- Surfaced term-based risk — 60-month loans warrant tighter approval controls and adjusted pricing
- Uncovered \$28.2M in permanent losses not visible from the charge-off count alone
- Built evidence for grade × purpose × term segmentation to inform approval policy and pricing strategy

Dashboard Overview

4-page Power BI dashboard, each page built around a single business question:

- **Page 1 — Summary:** Is the portfolio healthy? — KPI strip with MTD/MoM metrics, loan status breakdown, grade distribution, verification analysis, income band split
- **Page 2 — Risk View:** Where is the risk? — Dual-axis trend (volume vs charge-off rate), purpose chart colored by SQL risk tiers, term comparison, geographic map

- **Page 3 — Detail:** Show me individual loans. — 14-column row-level grid with Risk Tier on every loan, recovery panel locked to portfolio totals, 6 slicers
- **Page 4 — Insights:** What does it mean? — 8 insight cards, 5 recommendations, each traceable to a specific visual or filter combination

Data Modeling (SQL)

Normalized a 24-column flat CSV into a star schema — 1 fact table, 4 dimension tables:

- **Fact Table:** `fact_loan` — loan-level metrics with surrogate keys to all four dimensions
- **Dimensions:**
 - `dim_grade` — Grades A–G enriched with `risk_label` (Very Low Risk → Extreme Risk), surfaced in dashboard tooltips
 - `dim_purpose` — Loan purpose with `risk_tier` derived via SQL CTE from actual charge-off rates — High >18%, Medium 11–18%, Low <11%
 - `dim_date` — Full 365-day calendar via stored procedure; source had only 65 sparse dates, insufficient for time intelligence
 - `dim_region` — State codes mapped to US Census Bureau groups (Northeast / South / Midwest / West)
- `dim_borrower` evaluated and rejected — one row per loan = same granularity as the fact table; no repetition means no dimensional value
- `risk_tier` flows end-to-end: SQL CTE → `dim_purpose` → bar chart colors → Risk Tier column on every row in the detail grid

Technical Highlights

- **SQL · Risk classification:** CTE calculates charge-off rate per purpose and assigns tiers dynamically — no hardcoded labels
- **SQL · Key debugging:** `purpose_id` misaligned across all 14 purposes — `dim_purpose` loaded in charge-off order, `fact_loan` in source order; diagnosed, corrected, and validated
- **DAX · Time intelligence:** `CALCULATE + DATESMTD` used instead of `TOTALMTD` for averages — `TOTALMTD` is additive-only; average of monthly averages ≠ true period average
- **DAX · Bar colors from data:** A `SWITCH` measure returns hex strings from `dim_purpose[risk_tier]` — chart colors driven by the SQL model, not manual formatting
- **DAX · Recovery panel lock:** `CALCULATE` with a hardcoded 'Charged Off' filter preserves portfolio-level recovery totals regardless of active slicers
- **Validation:** Every KPI cross-checked against SQL ground truth before sign-off; rounding differences between 1dp and 2dp display documented

Notable Analytical Decisions

- **Recovery panel — self-initiated:** Not in the original scope. The \$28.2M net loss figure is more meaningful than a raw charge-off count and reframes how portfolio health is interpreted
- **Current loans shown separately in amber:** Not grouped with Fully Paid as 'Good Loans' — they haven't completed repayment and could still default. Silent assumptions don't belong in an analytical dashboard

What This Project Demonstrates

- **End-to-end ownership:** raw data → star schema → validated SQL → Power BI dashboard → business insights
- **SQL & data modeling:** schema design, CTE pipelines, stored procedures, surrogate key management
- **DAX proficiency:** time intelligence, filter context control, data-driven formatting
- **Analytical reasoning:** modeling decisions documented with logic — choices explained, not just executed
- **Business communication:** insights framed around decisions and financial impact, not just data observations